

JavaScript Challenge – 1

Flatten a Nested Array



Program :	JavaScript Interview Coding Challenges
Challenge No.	1
Challenge Title	Flatten a Nested Array
Main Concept	Recursion, Array

1. Objective

To understand and implement array flattening using recursion in JavaScript.

2. Problem Statement

Write a JavaScript function to flatten a nested array of any depth and return a new array with all the elements in a single level.

3. Requirements

1. Create a function named **flattenArray()** that takes an array as input.
2. The input array can contain elements of different data types (numbers, strings, booleans, etc.) and can be nested to any depth.
3. Use recursion to flatten the array.
4. Do not use the built-in `Array.flat()` method.
5. The function should return a new array with all the elements in a single level.
6. Add meaningful comments to explain the logic.

4. Sample Test Data and Expected Output

Sample Test Data (Input)	Expected Output
<code>flattenArray([1, [2, 3], [4, [5, 6]])</code>	<code>[1, 2, 3, 4, 5, 6]</code>
<code>flattenArray(["a", ["b", "c"], "d"])</code>	<code>["a", "b", "c", "d"]</code>
<code>flattenArray([1, [2, [3, [4]]], 5])</code>	<code>[1, 2, 3, 4, 5]</code>
<code>flattenArray([true, [false, [true]], "test"])</code>	<code>[true, false, true, "test"]</code>
<code>flattenArray([])</code>	<code>[]</code>

5. Technical Guidelines

- Use plain JavaScript (ES6 or later).
- Implement the solution using recursion.
- Do not use the built-in `Array.flat()` method.
- Handle arrays with elements of different data types.
- Ensure the original array is not modified.
- Add comments for better understanding.
- Follow proper naming conventions and code formatting.

6. Learning Outcome

After completing this challenge, you should be able to:

- Understand how to work with nested arrays in JavaScript.
- Implement recursion for solving complex problems.
- Write clean and reusable functions.
- Handle arrays with different data types and nested structures.